

Analysis of data with interpretable structure

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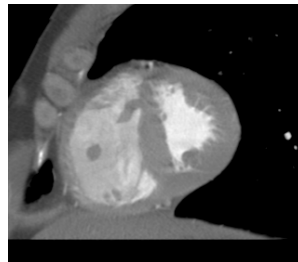
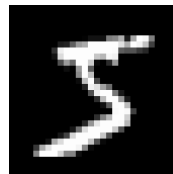
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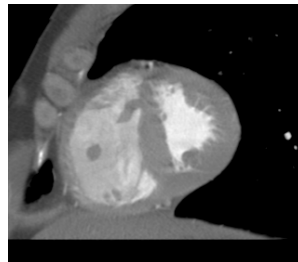
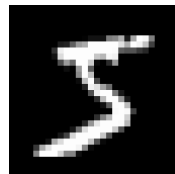
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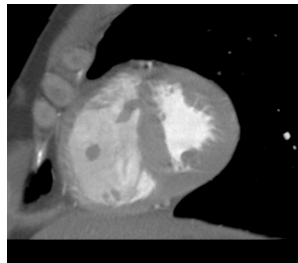
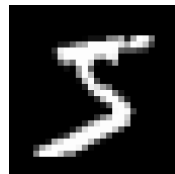
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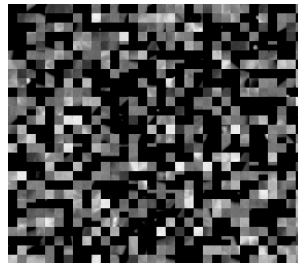
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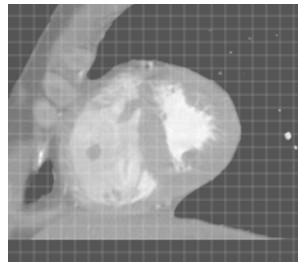
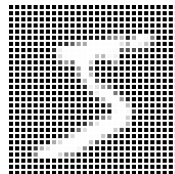
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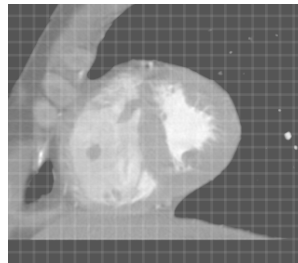
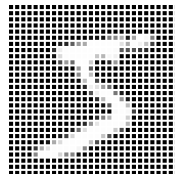
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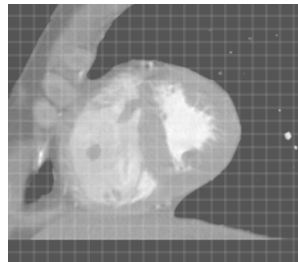
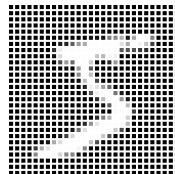
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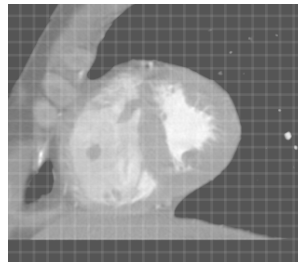
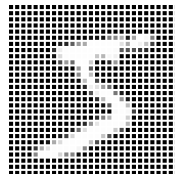
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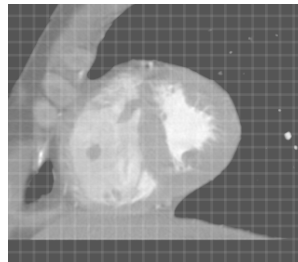
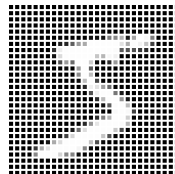
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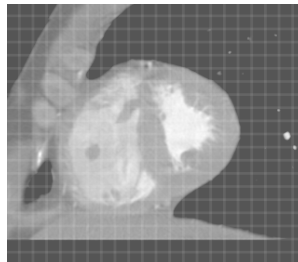
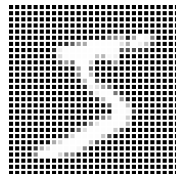
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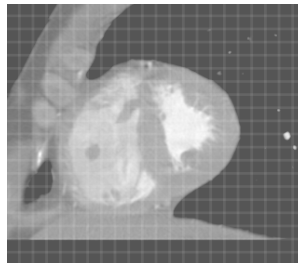
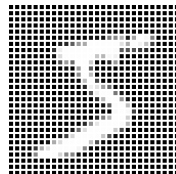
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- fortunately, we can change the internal representation of data

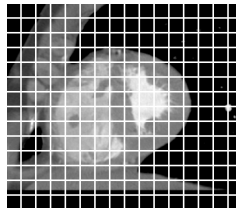
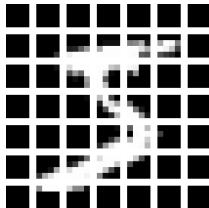


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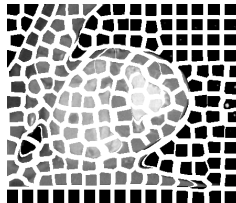
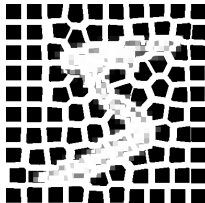
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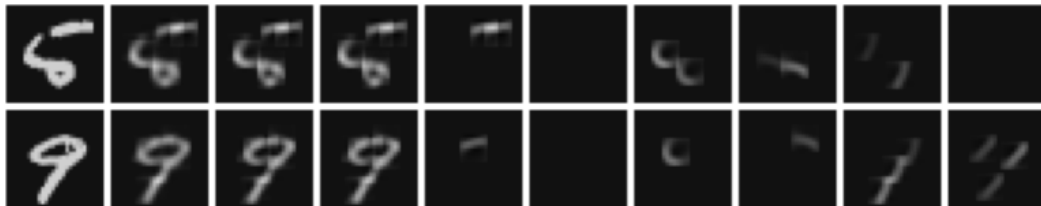
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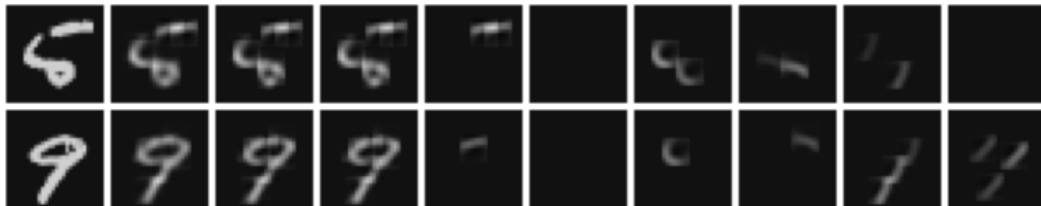


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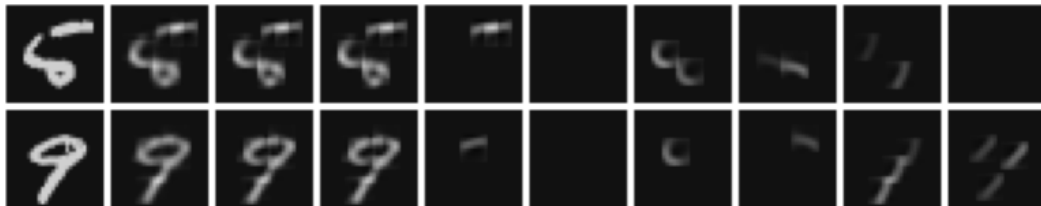
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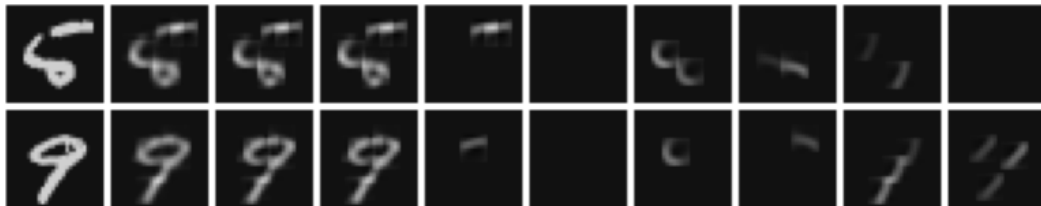
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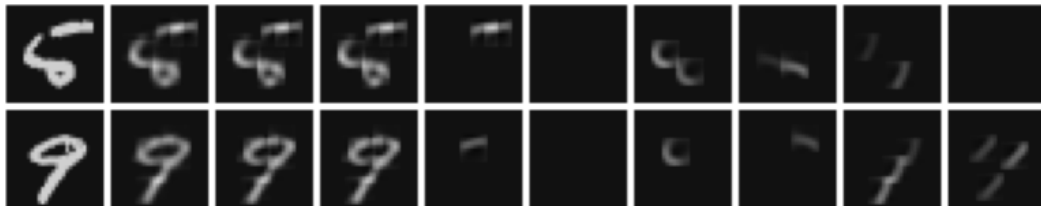
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- it would be optimal if the components were interpretable

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- depending on component and structure type, we can use different neural models to process them → *MLP, CNN, LSTM, Transformer, GNN* [4, 5, 6, 7]

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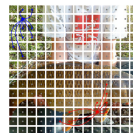
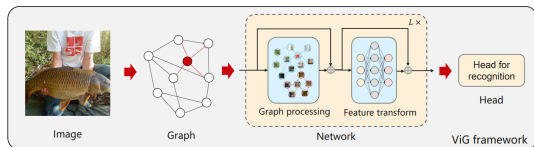
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Design architectures with interpretable working principles.

Design architecture that allows to discover important structure components.

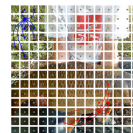
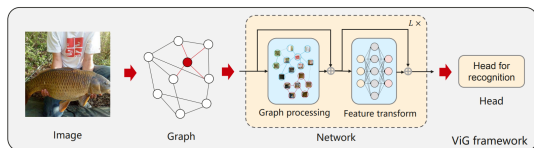
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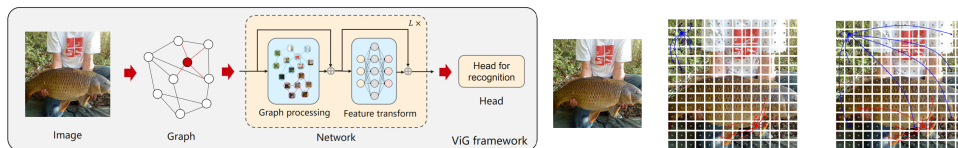
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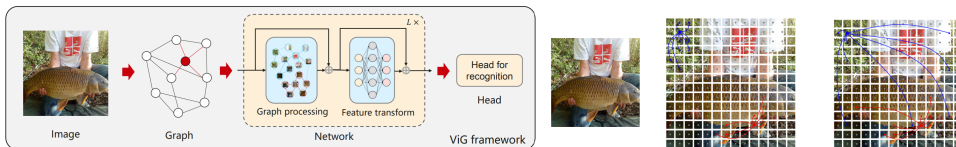
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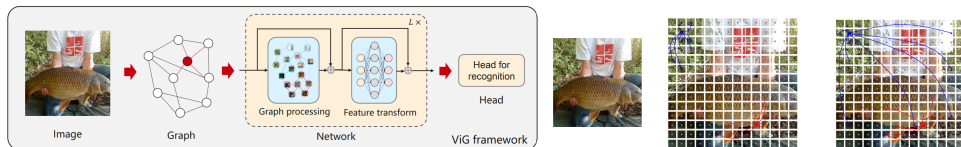
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 - it used k-NN to determine neighbour patches in the graph
 - the receptive field of each patch was much larger than in visualizations
 - the method allowed finding similar patches, but did not explain which patches were important for classification

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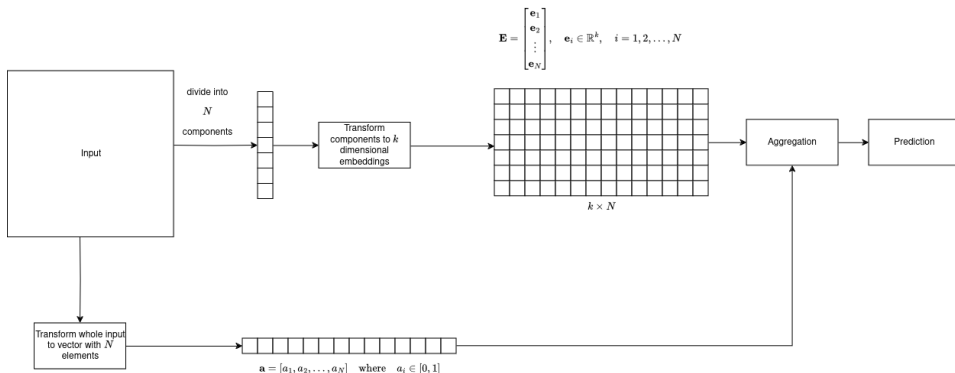
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 - replacing CNN based head with simple linear layers

Architecture

- these experiments allowed us to create general framework for building networks which can discover meaningful structure components in different domains and problems.

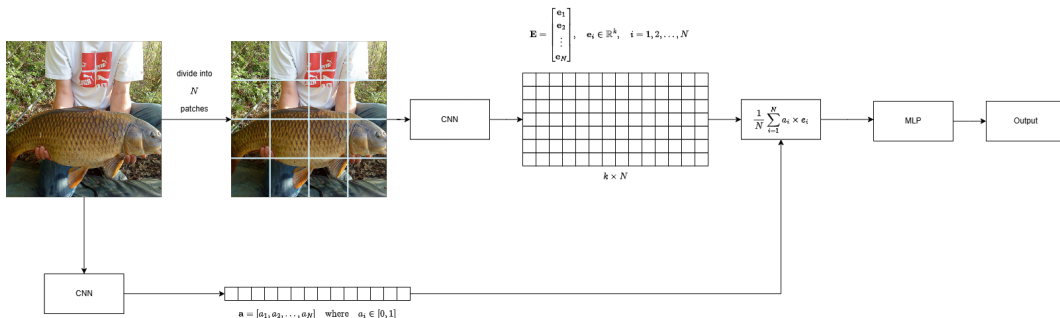
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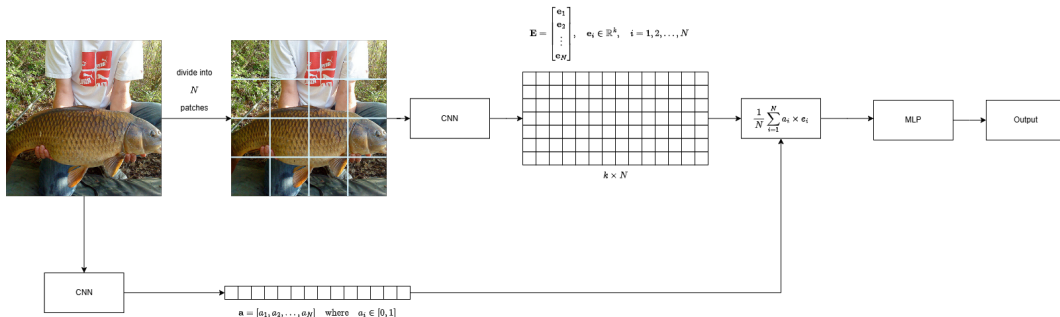
Approach

- this general framework can be applied for images split into components (patches)

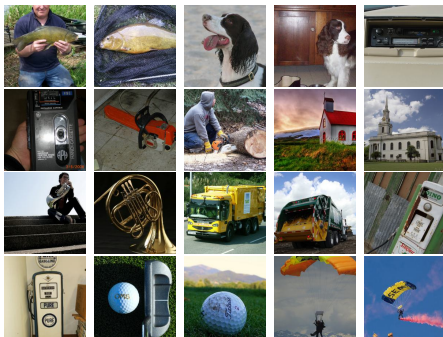


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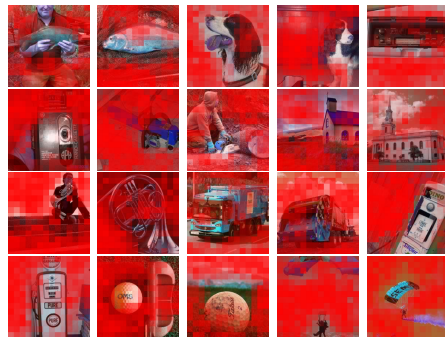
- this general framework can be applied for images split into components (patches)
- there is one CNN network creating N component embeddings of size k and second CNN network with N outputs that represent components importance



Results



(a) input images



(b) patch importances

Results

method	without importance		with importance	
	mean [%]	standard deviation [%]	mean [%]	standard deviation [%]
ours ($k = 2048$)	67.03	0.88	68.81	0.63
ours ($k = 1024$)	66.28	0.40	67.69	0.82
ours ($k = 512$)	65.33	1.28	66.30	1.49
ours ($k = 256$)	64.70	0.54	64.59	1.50
ours ($k = 128$)	62.63	1.01	62.56	1.01
ours ($k = 64$)	60.97	0.24	61.87	1.78
ours ($k = 8$)	54.25	0.59	54.01	1.31

Table: Test set classification evaluation (accuracy) for different embedding sizes k (statistics from 3 runs).

method	mean [%]	standard deviation [%]
ResNet-18	73.49	0.24

Results

method	mean [%]	standard deviation [%]	visible mask
ours CNN (freezed importance)	86.68	0.48	✓
ours CNN (unfreezed)	86.51	0.68	×
ours CNN (reinitialized)	85.92	0.26	×
ours MLP	76.49	1.12	✓

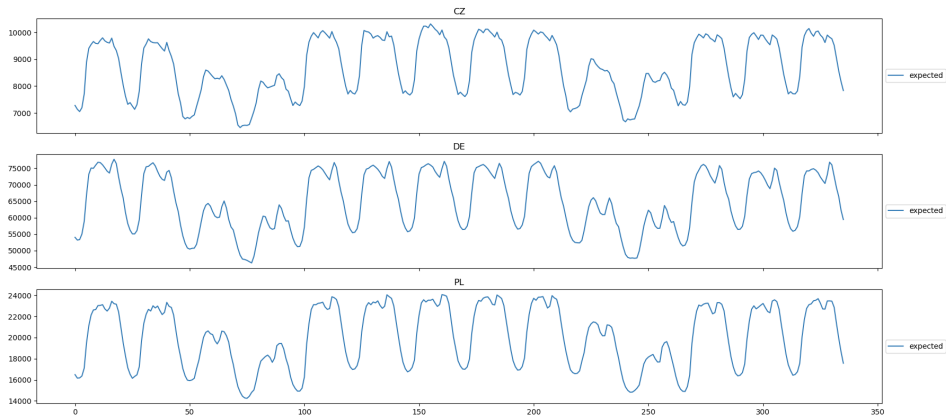
Table: Test set classification evaluation (accuracy) for our methods with augmentation (statistics from multiple runs).

method	mean [%]	standard deviation [%]	visible mask
ours CNN (freezed importance)	86.68	0.48	✓
ResNet-18	85.13	0.96	×
ResNet-50	83.33	0.92	×

Table: Comparison with state-of-the-art architectures.

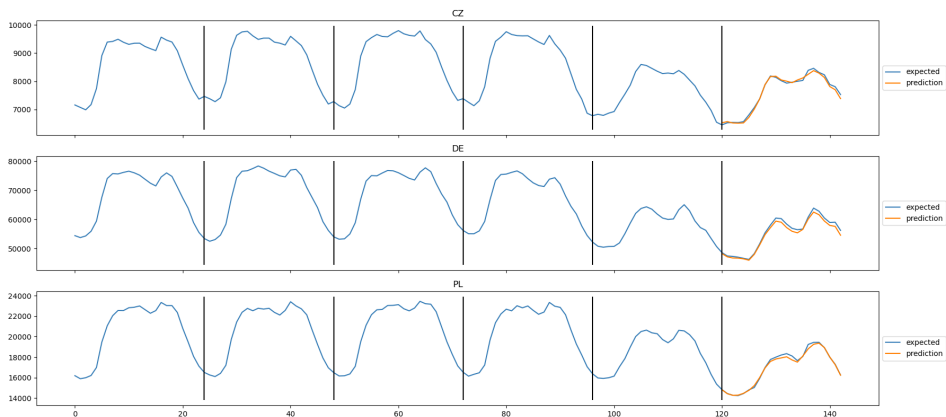
Data

- energy consumption in selected European countries measured once per hour



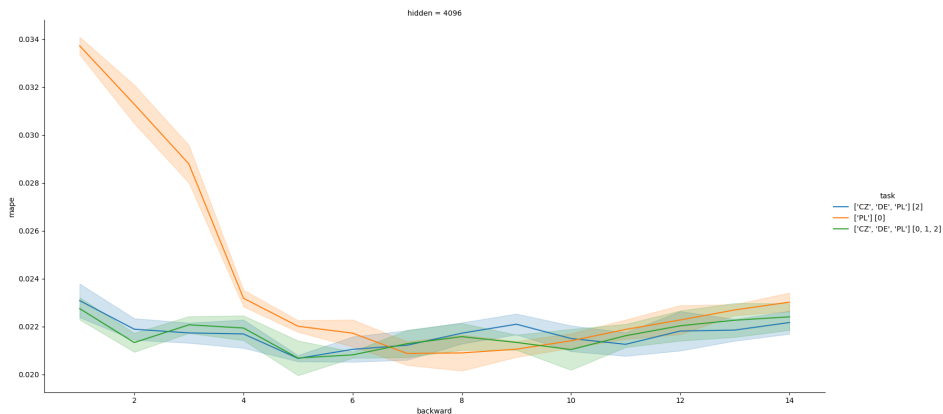
Task

- predict energy consumption for the next day basing on some number of backward days N



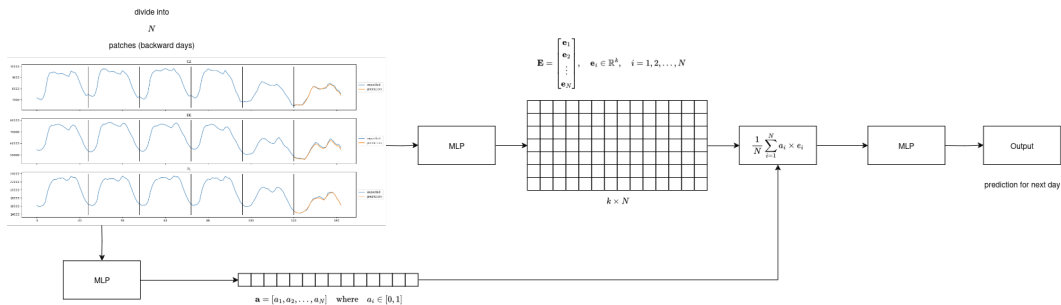
Reference

- we can do that using MLP network with $N \times 24$ inputs and 24 outputs



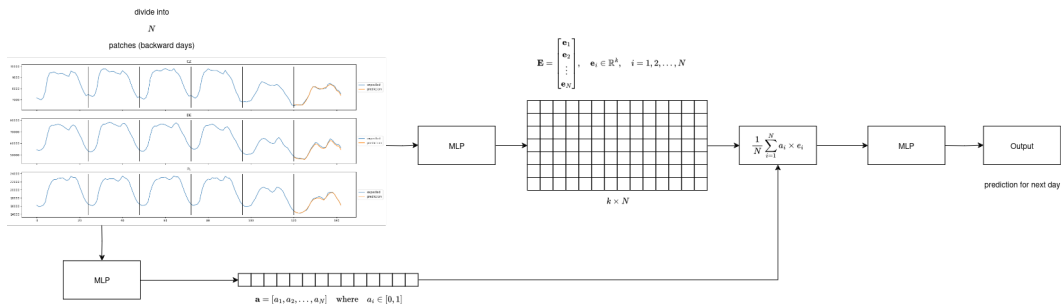
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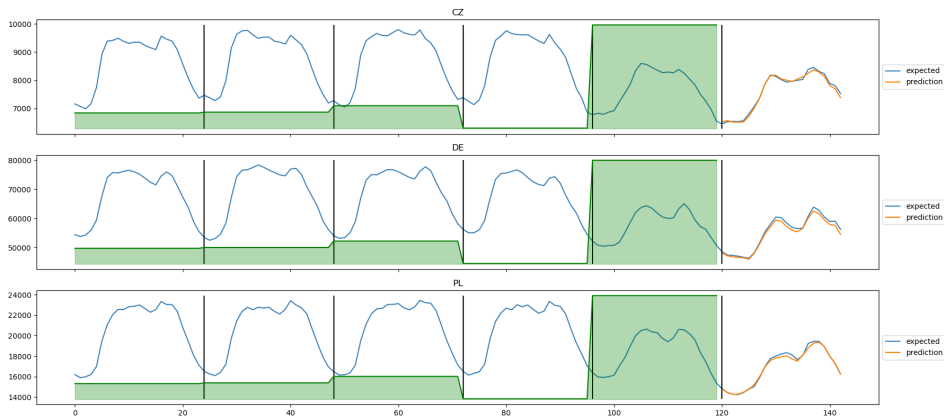
Approach

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- there is one MLP network with 24 inputs and k outputs creating component embeddings and second MLP network with $N \times 24$ inputs and N outputs that represent components importance

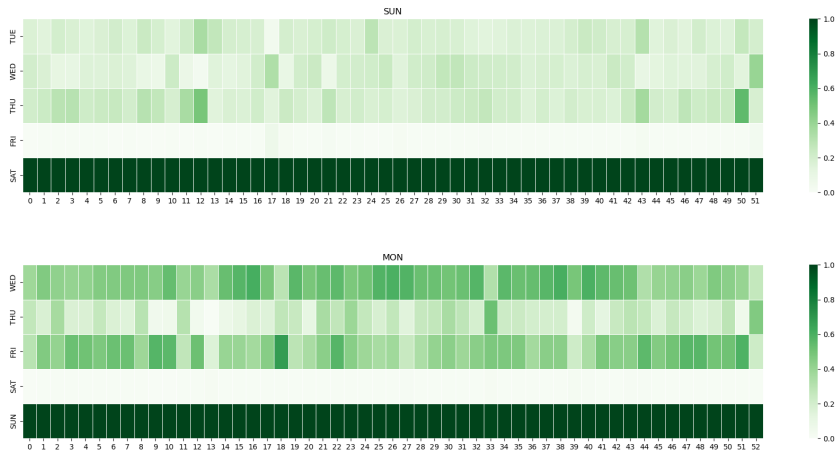


Results

- the proposed approach allows to discover, which backward days can have the highest influence on currently predicted day

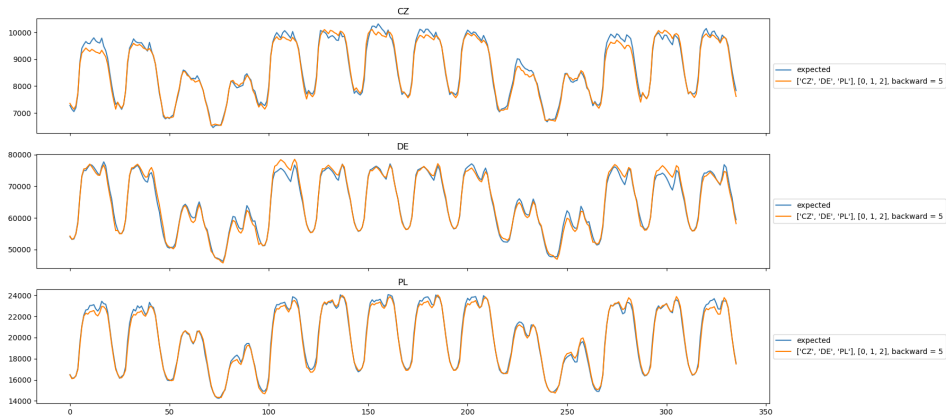


Results



- model on its own was able to discover specific patterns for every day of the week

Results



- although some components appeared to be less important prediction results were still satisfactory

Results

method	mean [%]	standard deviation [%]	min [%]
MLP ($N = 5$)	2.07	0.07	1.97
MLP ($N = 6$)	2.08	0.01	2.07
MLP ($N = 7$)	2.13	0.06	2.07
ours ($N = 5, k = 256$)	2.10	0.16	1.95
ours ($N = 5, k = 64$)	2.15	0.15	1.89
ours ($N = 6, k = 64$)	2.32	0.13	2.18
ours ($N = 7, k = 64$)	2.33	0.15	2.17
ours ($N = 6, k = 256$)	2.41	0.14	2.28
ours ($N = 7, k = 256$)	2.42	0.26	2.01

Table: Test set prediction evaluation (MAPE) for different architectures, component embedding sizes k , and backward days N (statistics from 5 runs).

- work on time series analysis is carried out in cooperation with prof. Grzegorz Dudek

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- future research directions:
 - more insight into current results

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Questions?

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